

Betty Wen

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PROFESSIONAL SUMMARY

University of Toronto

Sep 2022 – Present

Bachelor of Applied Science in Mechanical Engineering + PEY Co-op

Toronto, ON

Technical Skills: SolidWorks, AutoCAD, PDM, MBD, Autodesk, Python, ANSYS, COMSOL, LTSpice, MATLAB, Fusion 360, Minitab

Related Courses: Thermodynamics, Applied Fundamental of DeepLearning, HVAC Fundamentals

EXPERIENCE

Thermal and Mechanical Engineer

May 2025 – May 2026

Litens Automotive Group

Ontario, Canada

- Designed and optimized EV thermal components, with a focus on **3D printing** and **rapid prototyping**.
- Gained **hands-on experience** designing, assembling, and validating **nine 70:1 humanoid cycloidal actuators** across three design iterations, followed by torque-speed characterization and market benchmarking.
- Created **GD&T drawings** and integrated precision measurement data into 3D models for manufacturing and designs.
- Designed and built a **modular, reconfigurable** single-plate test rig with drive, sensor, and load modules for measuring torque-speed, and transmission efficiency under **3 N·m** deadweight calibration and **40 N·m** powered loads.
- Performed **FEA** under varying torque and load conditions to identify stress concentrations and recommend improvements to gear geometry, gear ratio, and material selection.

Mechanical Design Engineering Intern

May 2026 – August 2026

Tetrabot

China

- Developed the mechanical design and components for a mobile, rail-mounted EV-charging robot, with a primary focus on the guide-rail system for reliable motion and system integration.
- Developed a **spring-loaded traction** module using the existing mounting holes, integrating a rubber-coated bearing and approximately **30 N of preload** to improve **wheel-rail contact and reduce slipping**.
- Produced SolidWorks models, assemblies, system renderings, and manufacturing drawings. Applied hands-on knowledge of **sheet-metal fabrication, machining, and 3D printing** to create efficient and manufacturable designs.

Structural Engineer

June 2024 – August 2024

Yuwell Medical Devices

China

- Modeled and designed **vibration-damping** and **ventilation** components for oxygen concentrator compressors.
- Utilized 3D printing to prototype the thermal component and conducted performance tests on different positions of module, successfully maintaining the overall compressor temperature **below 55 Celsius degrees for over 1.5 hours**.
- Acquired **hands-on expertise** with oxygen concentrators, emphasizing the enhancement of component longevity, thermal efficiency, heat management, and manufacturability.

PROJECT

Condominium HVAC System Design

September 2026 – Present

University of Toronto

Toronto, Canada

- Designing a suite-based HVAC system to meet **ASHRAE 55 and 62.2** comfort and ventilation requirements, while considering humidity control, energy efficiency, noise, and maintainability.
- Calculating heating and cooling loads using the **CLTD/CLF method** and psychrometric analysis to determine ventilation and equipment requirements.
- Developing supply and exhaust duct layouts on **AutoCAD floor** plans using the equal-friction method; assessing diffuser placement, pressure drop, air distribution, and fan and coil selection.

Undergraduate Research Assistant

May 2024 - June 2024

NUS Multiphase Oil-Water-Air Flow Loop Laboratory, National University of Singapore

Singapore

- Conducted **CFD simulations** of gasoline-air multiphase flow using **ANSYS and COMSOL**, applying the **VOF method** to capture phase-interface evolution and to analyze pressure, velocity, and phase distributions.
- Performed FEA to evaluate pipeline stability and structural response, integrated the CFD and FEA findings, and authored a **12+ page research paper** proposing improvements to two-phase flow safety and efficiency.